# DISTRIBUTED LEDGER TECHNOLOGY

## Learning Outcome

- describe financial applications of distributed ledger technology

**Distributed ledger technology (DLT)** based on a distributed ledger (defined later) represents a technological development and offers potential improvements to delivering financing services and financial record keeping. DLT networks are being considered as a means to create, exchange, and track ownership of financial assets on a peer-to-peer (P2P) basis. Potential benefits of using this technology include greater accuracy, transparency, and security in record keeping; faster transfer of ownership; and P2P interactions. However, the technology is not fully secure, and breaches in privacy and data protection are possible. Additionally, the computational processes underlying DLT generally require massive amounts of energy to verify transaction activity.

A **distributed ledger** is a type of database that can be shared among potentially infinite numbers of entities in a network. In a distributed ledger, entries are recorded, stored, and distributed across a network of participants so that each participating entity has a matching copy of the digital database, making each copy of the database a verified record of all current and previous transactions. Basic elements of a DLT network include a digital ledger, a consensus mechanism used to confirm new entries, and a participant network.

The consensus mechanism is the process by which the computer entities (or nodes) in a network agree on a common state of the ledger. Consensus generally involves two steps: transaction validation and agreement on ledger update by network parties. These features enable the creation of records that are, for the most part, considered immutable, or unchangeable, yet they are transparent and accessible to network participants on a near-real-time basis. There are various approaches to establishing consensus.

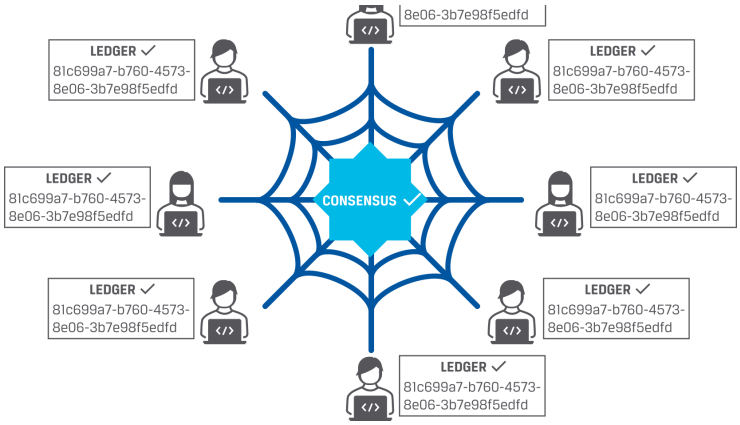
Features of DLT include the use of **cryptography**—an algorithmic process to encrypt data, making the data unusable if received by unauthorized parties—which enables a high level of network security and database integrity. For example, DLT uses cryptographic methods of proof to verify network participant identity and for data encryption.

DLT has the potential to accommodate "**smart contracts**," which are computer programs that self-execute on the basis of pre-specified terms and conditions agreed to by the parties to a contract. Examples of smart contract use are the automatic execution of contingent claims for derivatives and the instantaneous transfer of collateral in the event of default.

Exhibit 1 illustrates a distributed ledger network in which all participants (nodes) are connected to one another, each having a copy of the distributed ledger. The term "consensus" is shown in the center of the network and represents the consensus mechanism in which the nodes agree on new transactions and ledger updates.

**Exhibit 1:**

**Distributed Ledger Network Setup**

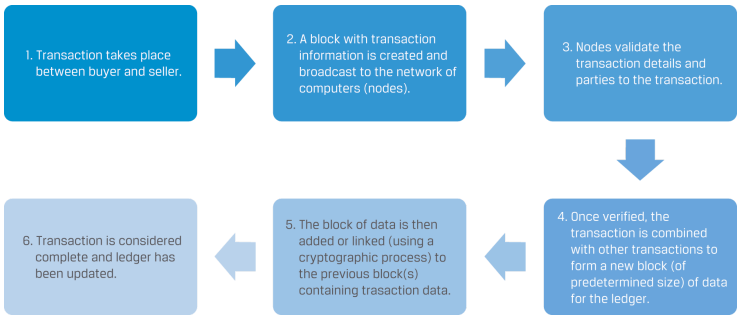



▶ Expand / collapse Extended Description

**Blockchain** is a type of digital ledger in which information, such as changes in ownership, is recorded sequentially within blocks that are then linked or "chained" together and secured using cryptographic methods. The steps outlining adding a new transaction to the network are outlined in Exhibit 2. There are several key steps a transaction must go through before it is added to the blockchain.

### Exhibit 2:

### Adding a Transaction to the Blockchain




▶ Expand / collapse Extended Description

Each block contains a grouping of transactions (or entries) and a secure link (known as a hash) to the previous block. New transactions are inserted into the chain only after validation via a consensus mechanism in which authorized members agree on the transaction and the preceding order, or history, in which previous transactions have occurred.

## Proof of Work vs. Proof of Stake

Fundamentally, blockchains are software protocols that enable many parties to interact under common assumptions and knowledge without having to trust each other. When transactions enter into a node of the blockchain, they are bundled into "blocks" and cryptographically "chained" together to facilitate verification of the prior history. How blocks are chained together is determined by the **consensus protocol**, a set of rules governing how blocks can join the chain and become the immutable "truth." The consensus protocols are designed to resist attempts at malicious manipulation up to a certain level of security. We distinguish between two main types of protocols.

## The Proof of Work (PoW) Protocol

The proof of work protocol determines which specific block to add through a computationally costly lottery. The PoW consensus mechanism used to verify a transaction involves a cryptographic problem that must be solved by some computers on the network (known as miners) each time a transaction takes place. **Miners** use powerful

themselves in the process. The “proof of work” consensus process to update the blockchain can require substantial amounts of computing power, making it very difficult and extremely expensive for an individual third party to manipulate historical data. To manipulate historical data, an individual or entity would have to control most nodes in the network. The success of the network, therefore, relies on broad network participation.

Participants on the network agree that the *longest chain* of blocks is the only truthful representation of all previous transactions. If a malicious attacker wanted to make a longer chain including fraudulent transactions, it must outperform the entire network’s computational power, which theoretically requires most of the network’s available computational power. This is the famous 51% attack threshold and serves as the boundary of the PoW protocol’s security standards. Amassing this much network power on the most widely used blockchains behind cryptocurrencies, such as Bitcoin or Ethereum, is extremely difficult for any individual, organization, or group of coordinated actors. Even if most of the network’s computational power can be controlled momentarily, the length of block history that can be altered is limited by the duration of this control. This has been the most widely used consensus protocol for digital assets.

## The Proof of Stake (PoS) Protocol

This protocol requires selected participants on the networks, the validators, to pledge capital to vouch for the block's validity. This stake signals to the network that a validator is available to verify the veracity of a transaction and propose a block. A majority of the other validators, who similarly staked a digital asset to the network, must then attest to the validity of proposed block. Validators benefit from both proposing and attesting to the validity of blocks that have been proposed by other participants in a similar staking process. The boundary of the PoS protocol’s security standards builds on a group of stakers (and their pledged stake) controlling the network’s computational power and protecting access from malicious parties gaining a majority.

Under both consensus protocols, the validation of the transactions comes with rewards: mining. Successful miners that validate the transaction obtain a new digital asset, whether it is a cryptocurrency or a token. The rate of mining and the speed at which new transactions on the network are validated generally control the amount of a new digital asset that can be created on a specific digital ledger network.

## Permissioned and Permissionless Networks

DLT can take the form of either permissionless or permissioned networks. **Permissionless networks** are open to any user who wishes to make a transaction, and all users within the network can see all transactions that exist on the blockchain. In a permissionless, or open, DLT system, any network participant can perform all network functions.

The main benefit of a permissionless network is that it does not depend on a centralized authority to confirm or deny the validity of transactions, because this takes place through the chosen consensus mechanism. This means no single point of failure exists because all transactions are recorded on a single distributed database and every node stores a copy of the database. Once a transaction has been added to the blockchain, it cannot be changed, barring manipulation; the distributed ledger becomes a permanent and immutable record of all previous transactions. In a permissionless network, trust between transacting parties is not a requirement.

**Bitcoin** is a well-known use of an open permissionless network. Bitcoin was created in 2009 to serve as the public ledger for all transactions occurring on its virtual currency. Since the introduction of Bitcoin, many more cryptocurrencies, or digital currencies, which use permissionless DLT networks, have been created.

In **permissioned networks**, network members might be restricted from participating in certain network activities. Controls, or permissions, can be used to allow varying levels of access to the ledger, from adding transactions (e.g., a participant) to viewing transactions

blockchains.

### Exhibit 3:

### Features of Permissioned and Permissionless Blockchains

| | Permissioned blockchain | Permissionless blockchain |
|---|---|---|
| **Speed** | Faster as only a limited number of members participate or are authorized to validate transactions | Slower as a large number of members have to reach consensus, which decreases network speed and scalability |
| **Cost** | Cost-effective as few members are required to validate each transaction | Not cost-effective as many members are required to validate each transaction |
| **Decentralization** | Partially decentralized as there are a limited number of members in the chain | Decentralized as all members can access the network |
| **Access** | Membership is limited. | Membership is unlimited. |
| **Governance** | The governance is determined by a centralized organization. | The governance is decentralized and is maintained by the members. |

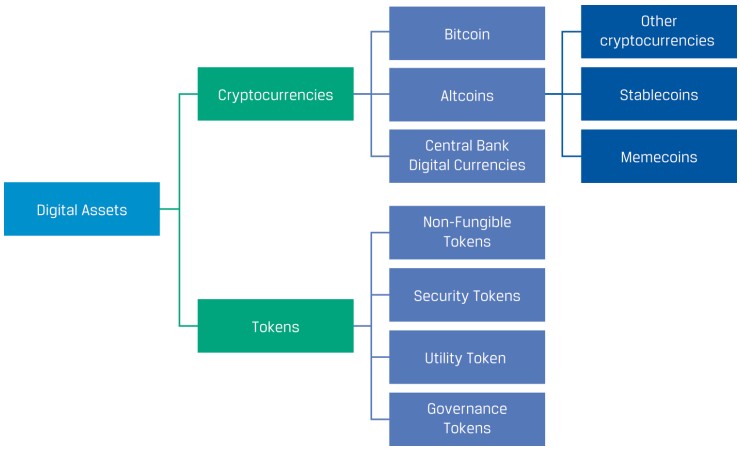
Based on their decentralized nature, permissionless distributed ledgers are fully open to all, but that high level of decentralization usually requires more processing power and bandwidth, making them less cost-effective. The opposite holds true for permissioned blockchains. Governance and transaction verification are two other aspects that are specific to the type of blockchain.

## Types of Digital Assets

Potential applications of DLT to financial services and investment management include the creation of digital assets, such as cryptocurrencies, and the process to deliver financial services more efficiently, such as processes tokenization, post-trade clearing and settlement, and compliance. Exhibit 4 summarizes different digital assets.

### Exhibit 4:

### Different Digital Assets

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## Cryptocurrencies

Digital assets are assets that exist only as an electronic record with rights to use, buy, or sell. They can be securities, currencies, properties, or commodities. The most common digital assets are units used to transfer or store value, referred to as **cryptocurrency** and

As electronic mediums of exchange, cryptocurrencies lack physical form and are issued privately by individuals, companies, and other organizations. Hence, they do not benefit from the backing of a central bank or a monetary authority. Most issued cryptocurrencies utilize open DLT systems in which a decentralized distributed ledger is used to record and verify all digital currency transactions. Several different types of cryptocurrency exist, such as altcoins and stablecoins.

Many cryptocurrencies have a self-imposed limit on the total amount of currency they may issue. Although such limits could help maintain their store of value, it is important to note that many cryptocurrencies have experienced high levels of price volatility. There is no consensus on how to value cryptocurrencies, and the apparent lack of clear fundamentals underlying these currencies has contributed to their volatility.

Cryptocurrencies are not government backed or regulated. Central banks around the world, however, are recognizing potential benefits and examining use cases for their own cryptocurrency versions as an alternative to physical currency. Such **central bank digital currencies (CBDCs)** are typically designed as a tokenized version of the currency issued by the central bank ("fiat currency")—essentially a digital bank note or coin.

## Tokens

Transactions involving physical assets, such as real estate, luxury goods, and commodities, often require substantial efforts in ownership verification and examination each time a transfer in ownership takes place. Through **tokenization**, the process of representing ownership rights to physical assets on a blockchain or distributed ledger, DLT has the potential to streamline this process by creating a single, digital record of ownership with which to verify ownership title and authenticity, including all historical activity.

Another form of digital assets is the **non-fungible token (NFT)**. An NFT links digital assets to certificates of authenticity using blockchain technology. NFTs differ from "fungible" tokens, such as cryptocurrencies, because each token and the authenticated object it represents is unique. As such, they uniquely "stamp" assets and can represent digital assets in a virtual world. The most common use for NFTs is trading in digital artwork.

**Security tokens** digitize the ownership rights associated with publicly traded securities. The custody of these security tokens can be stored on a blockchain, which increases the efficiency of post-trade processing, settlement, record keeping, and custody. Such a single ledger solution would eliminate the need for reconciliation and validation of transactions and allow participants to perform various transactions more easily and with greater transparency.

An example of security tokens is **initial coin offering** (ICO), an unregulated process whereby companies sell their crypto-tokens to investors in exchange for money or for another agreed upon cryptocurrency. An ICO is typically structured to issue digital tokens to investors that can be used to purchase future products or services being developed by the issuer. ICOs provide an alternative to traditional, regulated capital-raising processes, such as IPOs. Compared to the regulated IPO market, ICOs might have lower associated issuance costs and shorter capital-raising time frames. However, most ICOs do not typically have attached voting rights. Regulation for ICOs is under consideration in several jurisdictions, and numerous instances of investor loss have resulted from fraudulent schemes.

There are also **utility tokens** that provide services within a network, such as pay for services and network fees. While security tokens may pay out dividends, utility tokens only compensate for activities on the network.

**Governance tokens** are important in permissionless networks. They serve as votes to determine how particular networks are run. For instance, when there is a technical problem on a permissionless blockchain network, it is the owners of governance tokens

## Question Set

**1.** Which of the following is *not* a potential benefit of distributed ledger technology?

**A.** Facilitation of smart contracts
**B.** Energy-efficient way of record keeping
**C.** Immutable and secure transaction records

Hide Solution

**Solution:**

The correct answer is B. Distributed ledger technology provides greater accuracy, transparency, and security in record keeping; enables faster transfer of ownership; and enables peer-to-peer interactions. A DLT network relies on certain consensus mechanisms in which all nodes that are connected on the network agree on new transactions and ledger updates. Once verified by all the nodes of the network, the transaction ledger is immutable and is kept by each of the nodes. However, the transaction validation process requires material computational power of all the miners on the network, especially in the case of proof-of-work consensus protocol.

**2.** Identify the following statement as true or false: Prices of cryptocurrencies can be volatile because there are finite limits on the total amount of currencies that may be issued.

Hide Solution

**Solution:**

False. Prices of cryptocurrencies have been volatile because there is no consensus on how to value cryptocurrencies. The apparent lack of clear fundamentals underlying these currencies has contributed to their volatility.

**3.** The process where a node on a blockchain network pledges its digital asset to verify a new block's validity is called:

**A.** tokenization.
**B.** proof of work.
**C.** proof of stake.

Hide Solution

**Solution:**

The correct answer is C. This proof-of-stake protocol requires selected participants on a blockchain network, the validators, to pledge digital assets to vouch for the block's validity. This stake signals to the network that a validator is available to verify the veracity of a transaction and propose a block. Other validators who stake a digital asset to the network must then attest to the validity of proposed block. Validators benefit from both proposing and attesting to the validity of blocks that have been proposed by other participants in a similar staking process in the form of new digital assets.

**4.** Identify the various types of digital assets according to their general classifications:

| | |
|---|---|
| A. Cryptocurrencies | 1. ICO |
| B. Tokens | 2. NFT |
| | 3. CBDC |
| | 4. Ether |

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**Solution:**

| | |
|---|---|
| A. Cryptocurrencies | 3. CBDC |
| | 4. Ether |
| | 5. Stablecoin |
| B. Tokens | 1. ICO |
| | 2. NFT |
| | 6. Utility token |

Discuss Filter

Discussion

What do you want to discuss... (required)

PE "the computational processes underlying DLT generally require massive amounts of energy to verify transaction activity." is not necessarily correct... It depends on the consensus protocol. Bitcoin cannot change its PoW consenssus protocol. And PoW is energy intensive.
But other blockchains run on different protocols, or changed theirs. Like ethereum, when it changed to PoS, barely consuming energy (aprox same as the energy consummed by posting 1 comment on X). So I don't think that it is correct to say that computational processess underlying DLT technology GENERALLY require massive amounts of energy to verify transactions. It only does when the protocol on which it runs is energy intensive - like PoW or similars- Bitcoin, litecoin, dogecoin, monero... And it doesn t when it s not (PoS and similars)- Ethereum (after Merge), Cardano, Solana, Polkadot, EOS, TRON, Steem, hyperledger, Avalanche...

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